

Mixture and Alligation

Mixture: When two or more than two substances are mixed in any ratio to produce a product is known as Mixture.

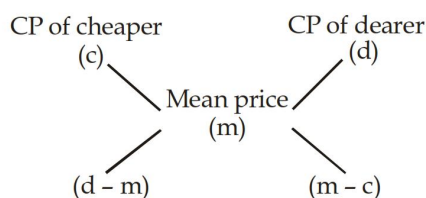
Mean price: The cost price of a unit quantity of the mixture is called the mean price.

Concept 1. Alligation Rule

Let the cost price of a unit of cheaper article is Rs. c and that of a unit of costly article is d and the average (mean) price of mixture is m , then

$$\frac{\text{Quantity of cheaper article}}{\text{Quantity of costly article}} = \frac{\text{Cost price of a unit of costly article} - \text{Mean price}}{\text{Mean price} - \text{Cost price of a unit of a cheaper article}}$$

The above relation is represented as

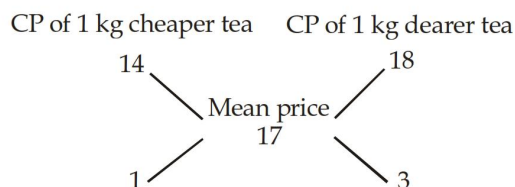


Hence,
$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = (d - m) : (m - c) = \frac{d - m}{m - c}$$

Alligation rule is also used to find the ratio in which two or more ingredients at their respective prices should be mixed to produce a mixture at a given price.

Example: In what proportion must tea at Rs. 14 per kg be mixed with tea at Rs. 18 per kg, so that the mixture be worth Rs. 17 a kg?

Sol.



$$\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = \frac{18 - 17}{17 - 14} = \frac{1}{3} = 1 : 3$$

Concept 2. A container has milk and water in the ratio $a:b$, a second container of same capacity as first one has milk and water in the ratio $c:d$. If both the mixture are emptied into a third container, then the ratio of milk to water in third container is given by

$$\left[\frac{a}{a+b} + \frac{c}{c+d} \right] : \left[\frac{b}{a+b} + \frac{d}{c+d} \right]$$

Example: There are two containers of equal capacity. The ratio of milk to water in the first container is $3 : 1$, in the second container is $5 : 2$. If they are mixed up, then the ratio of milk to water in the mixture will be?

Sol. Part of milk in first container $= \frac{3}{3+1} = \frac{3}{4}$

Part of water in first container $= \frac{1}{3+1} = \frac{1}{4}$

Similarly, part of milk in second container = $\frac{5}{5+2} = \frac{5}{7}$

Part of water in second container = $\frac{2}{5+2} = \frac{2}{7}$

$$\therefore \text{Required} = \frac{3}{4} + \frac{5}{7} : \frac{1}{4} + \frac{2}{7} = \frac{41}{28} : \frac{15}{28} = 41 : 15$$

Concept 3: Suppose a container contains x units of a liquid from which y units are taken out and replaced by water.

After n operation, quantity of pure liquid = $x \left(1 - \frac{y}{x}\right)^n$ units

Example: A container contains 40 litres of milk. From this container 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?

Sol. Amount of milk left after 3 operations = $\left[40 \left(1 - \frac{4}{40}\right)^3\right]$ litres = $\left(40 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10}\right) = 29.16$ litres.

Types of Questions

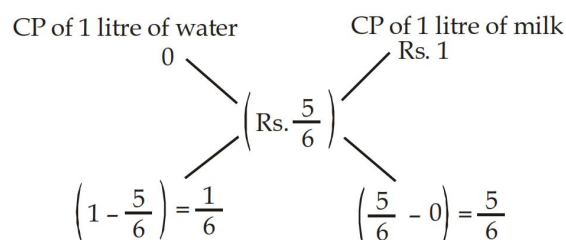
1. In what ratio must water be mixed with milk to gain 20% by selling the mixture at cost price?

Sol. Let CP of milk be Rs. 1 per litre.

Then; SP of 1 litre of mixture = Rs. 1, Gain = 20%.

$$\therefore \text{CP of 1 litre of mixture} = \text{Rs.} \left(\frac{100}{120} \times 1\right) = \text{Rs.} \frac{5}{6}.$$

By the rule of alligation, we have:

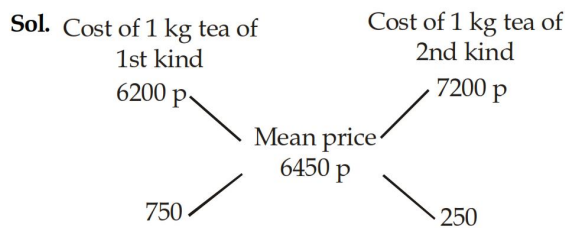


$$\therefore \text{Ratio of water and milk} = \frac{1}{6} : \frac{5}{6} = 1 : 5.$$

Shortcut:

$$\text{Ratio} = \text{Profit}\% : 100 = 20 : 100 = 1 : 5$$

2. In what ratio must tea at Rs. 62 per kg be mixed with tea at Rs. 72 per kg so that the mixture must be worth Rs. 64.50 per kg?



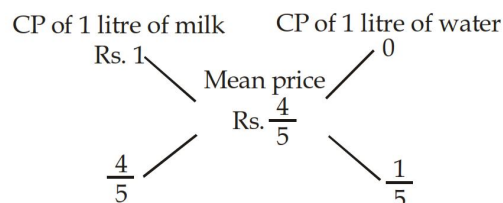
$$\therefore \text{Required ratio} = 750 : 250 = 3 : 1.$$

3. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is:

Sol. Let CP of 1 litre milk be Rs. 1.

Then, SP of 1 litre of mixture = Rs. 1, Gain = 25%.

$$\text{CP of 1 litre mixture} = \text{Rs.} \left(\frac{100}{125} \times 1\right) = \text{Rs.} \frac{4}{5}.$$



$$\therefore \text{Ratio of milk to water} = \frac{4}{5} : \frac{1}{5} = 4 : 1.$$

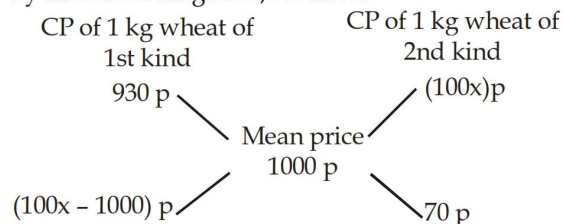
Hence, percentage of water in the mixture

$$= \left(\frac{1}{5} \times 100\right)\% = 20\%.$$

4. One quality of wheat at Rs. 9.30 per kg is mixed with another quality at a certain rate in the ratio 8:7. If the mixture so formed be worth Rs. 10 per kg, what is the rate per kg of the second quality of wheat?

Sol. Let the rate of the second quality be Rs. x per kg.

By the rule of alligation, we have:



$$\therefore \frac{100x - 1000}{70} = \frac{8}{7} \Rightarrow 700x - 7000 = 560$$

$$700x = 7560 \Rightarrow x = \text{Rs. } 10.80.$$

5. A container contains 100 litres of spirit. From this container 10 litres of spirit was taken out and replaced by water. This process was repeated further two times. How much spirit is now contained by the container?

Sol. Amount of milk left after 3 operations

$$= \left[100 \left(1 - \frac{10}{100} \right)^3 \right] \text{ litres}$$

$$= \left(100 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10} \right) = 72.9 \text{ litres.}$$

Foundation

Questions

1. In a mixture of milk and water of the volume of 60 litre the ratio of milk to water is 7 : 5. How much quantity of water will be added to make mixture of equal ratio?
(a) 8 Litre (b) 10 Litre
(c) 15 Litre (d) 20 Litre
2. In a mixture of milk and water of the volume of 30 Litre, the ratio of milk and water is 8 : 7. How much water should be added in mixture to make ratio 4 : 5?
(a) 6 Litre (b) 4.5 Litre
(c) 5 Litre (d) 5.5 Litre
3. In what ratio must oil at Rs. 62 per kg be mixed with oil of Rs. 72 per kg, so that the mixture must be worth Rs. 64.50 per Kg?
(a) 1 : 3 (b) 1 : 2
(c) 3 : 1 (d) 4 : 1
4. A mixture of milk and water measures 60 ltr. It contains 20% water. How many litres of water should be added to it so that water may be 25%?
(a) 6 ltr (b) 4 ltr
(c) 8 gallons (d) 10 ltr
5. In what ratio must a grocer mix two varieties of pulses costing Rs. 15 and Rs. 20 per kg respectively so as to get a mixture worth Rs. 16.5 per kg?
(a) 3 : 7 (b) 5 : 7
(c) 7 : 3 (d) 7 : 5
6. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 20%. The percentage of water in the mixture is?
(a) 4% (b) 16.66%
(c) 20% (d) 25%
7. Find the ratio in which rice at Rs. 7.20 a kg be mixed with rice at Rs. 5.70 a kg to produce a mixture worth Rs. 6.30 a kg?
(a) 1 : 3 (b) 2 : 3
(c) 3 : 4 (d) 4 : 5
8. Aditya and Sanjay started a business investing 45000 and 30000 respectively. What will be the ratio between their profit?
(a) 2 : 3 (b) 3 : 2
(c) 4 : 9 (d) Can be determined
9. In what ratio must sugar at Rs. 2 per kg be mixed with sugar at Rs. 3.50 per kg so that the mixture be worth Rs. 2.50 per kg?
(a) 2 : 1 (b) 1 : 5
(c) 3 : 1 (d) 2 : 3
10. 600 gm of Sugar solution has 40% sugar in it. How much sugar should be added to make it 50% in the solution?
(a) 60 gm. (b) 90 gm.
(c) 120 gm. (d) 150 gm.
11. Gold is 21 times heavy as compared to water and copper is 11 times heavy as compared to water. In what ratio should these metal be mixed so that the mixture may be 17 times as heavy as water?
(a) 1 : 2 (b) 2 : 1
(c) 3 : 1 (d) 3 : 2
12. I have 100 books of English. I sold some of these at a profit of 12% and rest at 8% loss. On the whole I got a profit of 11%. How many books were sold at 12% profit by me?
(a) 50 (b) 95
(c) 57 (d) 60
13. There are 3 tub which contains mixtures of Milk and Water in the ratio of 5 : 2, 4 : 3 and 3 : 1 respectively. If the mixtures be poured in a single tub. Find the Ratio Milk and Water?
(a) 13 : 3 (b) 19 : 9
(c) 17 : 9 (d) Can't be determine
14. In what proportion must be Sugar of Rs. 17 per kg mixed with Sugar of Rs. 29 per kg to make mixture of Rs. 20 per kg?
(a) 2 : 3 (b) 17 : 29
(c) 29 : 17 (d) 3 : 1
15. In what proportion must pulse of Rs. 70 per kg be mixed with pulse of Rs. 45 per kg to get a mixture of Rs. 60 per Kg?
(a) 9 : 13 (b) 13 : 9
(c) 14 : 9 (d) 3 : 2

16. A shopkeeper has Rs. 50 kg of rice, of part of which he sells at 10% profit and the rest at 5% loss. He gains 7% on the whole. What is the quantity sold at 10% gain?
 (a) 50 (b) 40
 (c) 10 (d) Can't be determined
17. A man buys 2 Cow for Rs. 2700 and sells one for loss of 6% and on other he gains 7.5%. On the whole he neither gained nor lost. What does 2 nd Cow cost?
 (a) 1100 (b) 1000
 (c) 1200 (d) 6075
18. Aditya, Sonal and Nutan contributed Rs. 50,000 for business. Aditya contributed Rs. 4000 more than Sonal and Sonal contributed Rs. 5,000 more than Nutan. Out of total profit of Rs. 70,000 how much did Aditya receive?
 (a) Rs. 16,000 (b) Rs. 18,000
 (c) Rs. 20,000 (d) Rs. 29400
19. The ratio in which two sugar solutions of the concentrations 15% and 40% are to be mixed to get a solution of concentration 30% is:
 (a) 2 : 3 (b) 3 : 2
 (c) 8 : 9 (d) 9 : 8
20. Nikita bought 30 kg of wheat at the rate of Rs. 9.50 per kg and 40 kg of wheat at the rate of Rs. 8.50 per kg and mixed them. She sold the mixture at the rate of Rs. 8.90 per kg. Her total profit or loss in the transaction was:
 (a) Rs. 2 loss (b) Rs. 2 profit
 (c) Rs. 7 loss (d) Rs. 7 profit
21. A mixture of a certain quantity of milk with 16 litres of water is worth 90 P per litre. If pure milk be worth Rs. 1.08 per litre. How much milk is there in the mixture?
 (a) 40 litres (b) 50 litres
 (c) 60 litres (d) 80 litres
22. A person has a chemical of Rs. 25 per litre. In what ratio should water be mixed in the mixture so that by selling it at Rs. 20 per litre he may get a profit of 25%?
 (a) 14 : 9 (b) 16 : 9
 (c) 9 : 14 (d) 9 : 16
23. A mixture of 40 litres of milk and water contains 10% water. How much water must be added to make 20% water in the new mixture?
 (a) 3 litres (b) 4 litres
 (c) 5 litres (d) 6 litres
24. If 2 kg of metal, of which $\frac{1}{3}$ is zinc and the rest is copper, be mixed with 3 kg of metal, of which $\frac{1}{4}$ is zinc and the rest is copper, what is the ratio of zinc to copper in the mixture?
 (a) 13 : 42 (b) 17 : 43
 (c) 19 : 43 (d) 15 : 42
25. 50 g of an alloy of gold and silver contains 80% gold (by weight). The quantity of gold, that is to be mixed up with this alloy, so that it may contain 95% gold, is:
 (a) 200 g (b) 150 g
 (c) 50 g (d) 10 g
26. A sum of Rs. 6.40 is made up of 80 coins which are either 10-paise or 5-paise coins. How many are coins of 5-paise are there?
 (a) 24 (b) 28
 (c) 32 (d) 36
27. How much tea at Rs. 4 a kg should be added to 15 kg of tea at Rs. 10 per kg so that the mixture be worth Rs. 6.50 a kg?
 (a) 15 kg (b) 35 kg
 (c) 25 kg (d) 21 kg
28. Gold is 19 times as heavy as water and copper is 9 times as heavy as water. In what ratio should these metals be mixed so that the mixture may be 15 times as heavy as water?
 (a) 2 : 3 (b) 3 : 2
 (c) 2 : 4 (d) 4 : 2
29. A pot contains 81 litres of pure milk. $\frac{1}{3}$ of the milk is replaced by the same amount of water. Again $\frac{1}{3}$ of the mixture is replaced by that amount of water. The ratio of the milk and water in the new mixture is?
 (a) 1 : 2 (b) 1 : 1
 (c) 2 : 1 (d) 4 : 5
30. A man has 60 pens. He sells some of these at a profit of 12% and the rest at 8% loss. On the whole, he gets a profit of 11%. How many pens were sold at 12% profit:
 (a) 47 (b) 52
 (c) 55 (d) 57

Moderate

1. In what ratio must water be mixed with milk to gain 16% on selling the mixture at cost price?
 - (a) 1 : 6
 - (b) 4 : 25
 - (c) 2 : 3
 - (d) 4 : 3
2. In what ratio must a grocer mix two varieties of tea worth Rs. 58 per kg and Rs. 67 per kg so that by selling the mixture at Rs. 68.20 per kg he may gain 10%?
 - (a) 3 : 2
 - (b) 3 : 4
 - (c) 3 : 5
 - (d) 5 : 4
3. A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the containers so as to get 12 litres of milk such that the ratio of water to milk is 3 : 5?
 - (a) 4 litres, 8 litres
 - (b) 6 litres, 6 litres
 - (c) 5 litres, 7 litres
 - (d) 7 litres, 5 litres
4. How many kilogram of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per kg so that there may be a gain of 10% by selling the mixture at Rs. 9.24 per kg?
 - (a) 36 kg
 - (b) 42 kg
 - (c) 54 kg
 - (d) 63 kg
5. A container contains 50 litres of milk. From this container 5 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?
 - (a) 26.34 litres
 - (b) 27.36 litres
 - (c) 28 litres
 - (d) 36.45 litres
6. In a mixture of milk and water of Volume 30 Litre the ratio of water and milk is 3 : 7. How much quantity of water will be added to the mixture to make the ratio of milk and water 1 : 2
 - (a) 23 Litre
 - (b) 21 Litre
 - (c) 12 Litre
 - (d) 33 Litre
7. The cost of Type 1 rice is Rs. 15 per kg and Type 2 rice is Rs. 20 per kg. If both Type 1 and Type 2 are mixed in the ratio of 2 : 3, then the price per kg of the mixed variety of rice is?
 - (a) Rs. 18
 - (b) Rs. 18.50
 - (c) Rs. 19
 - (d) Rs. 19.50
8. A merchant has 1000 kg of sugar, part of which he sells at 8% profit and the rest at 18% profit. If he gains 14% on the whole then the quantity sold at 18% profit is:
 - (a) 400 kg
 - (b) 560 kg
 - (c) 600 kg
 - (d) 640 kg
9. In an Examination out of 900 students 85% of the boys and 70% of girls passed. How many girls appeared in the examination, if total pass percentage was 75%?
 - (a) 400
 - (b) 500
 - (c) 600
 - (d) 700
10. 400 gm of salt solution has 40% salt in it. How much salt should be added to make it 50% in the solution?
 - (a) 60
 - (b) 70
 - (c) 65
 - (d) 80
11. Chocolates at Rs. 12.00 per dozen is mixed with chocolate at Rs. 10.00 per dozen in the ratio 3 : 5. Find the price per dozen of the mixture?
 - (a) Rs. 11.00
 - (b) Rs. 9.90
 - (c) Rs. 11.50
 - (d) Rs. 10.75
12. Pencils at Rs. 4.20 per dozen is mixed with pencils at Rs. 5.40 per dozen in the ratio 3 : 5. Find the price per dozen of the mixture?
 - (a) Rs. 4.95
 - (b) Rs. 4.50
 - (c) Rs. 5.00
 - (d) Rs. 5.05
13. A mixture of 48 litre of milk and water contains 10% of water. How much water must be added to make 20% water in the new mixture?
 - (a) 2
 - (b) 6
 - (c) 4
 - (d) 5
14. A vessel of 160 litre is filled with Milk and Water. 70% of Milk and 30% of Water is taken out of the vessel. After this task, it is found that vessel is now filled by 55% quantity of Milk and Water. What is the original quantity of milk and water in the vessel respectively?
 - (a) 60, 100
 - (b) 100, 60
 - (c) 70, 90
 - (d) 90, 70
15. A mixture of 80 litre of milk and water contains 10% water. How much water must be added to make 20% water in the new mixture?
 - (a) 5 litre
 - (b) 8 litre
 - (c) 10 litre
 - (d) 15 litre
16. A tub contains a mixture of two liquid P and Q in the ratio of 4 : 1. When 10 Litre of the mixture is taken out and 10 Litre of Liquid Q is poured in the jar, the ratio becomes 2 : 3. How many litre of liquid P was contained in the jar?
 - (a) 10 litre
 - (b) 15 litre
 - (c) 20 litre
 - (d) 16 litre
17. A bottle contains 27 litres of pure milk. $\frac{1}{3}$ of the milk is replaced by the same amount of water. Again $\frac{1}{3}$ of the mixture is replaced by that amount of water. The ratio of the milk and water in the new mixture is:

- (a) 1 : 2 (b) 1 : 1
(c) 2 : 1 (d) 4 : 5
18. Aditya and Manish continued in a joint business for 36 months. Aditya contribute Rs. 300 for certain time and Manish invested Rs. 500 for remaining time. If out of total profit of Rs. 1020, Aditya gets Rs. 495, then for how long Aditya kept his money?
(a) 16 months (b) 14 months
(c) 8 months (d) 22 months
19. A jar contained a mixture of two liquids A and B in the ratio 3 : 2. When 5 litres of the mixture was taken out and 5 litres of liquid B was poured in the jar, this ratio became 2 : 3. The quantity of liquid A contained in the jar initially was:
(a) 4 litres (b) 8 litres
(c) 9 litres (d) 32 litres
20. A man has 40 kg of tea, a part of which he sells at 5% loss and the rest at the cost price. In this business he incurs a loss 3%. Find the quantity which he sells at the cost price?
(a) 12 kg (b) 14 kg
(c) 16 kg (d) 18 kg
21. In a mixture of 75 litres, the ratio of milk to water is 2:1. The amount of water to be further added to the mixture so as to make the ratio of the milk to water 1 : 2 will be:
(a) 45 litres (b) 60 litres
(c) 75 litres (d) 80 litres
22. How many kg. of salt at 42 P per kg. must a man mix with 25 kg of salt at 24 P per kg, so that he may, on selling the mixture at 40 P per kg, gain 25% on the outlay?
(a) 20 (b) 30
(c) 40 (d) 50
23. 300 gm of sugar solution has 40% sugar in it. How much. sugar should be added to make it 50% in the solution?
(a) 40 gm (b) 50 gm
(c) 60 gm (d) 80 gm
24. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 40 % The percentage of water in the mixture is:
(a) 25 % (b) 20 %
(c) 45 % (d) 28.56 %
25. 200 litres of mixture contains 15% water and the rest is milk. The amount of milk that must be added so that the resulting mixture contains 87.5% milk is:
(a) 30 litre (b) 35 litre
(c) 40 litre (d) 45 litre
26. In a zoo, there are rabbits and pigeons. If the heads are counted, there are 200 and if legs are counted, there are 580. How many pigeons are there?
(a) 105 (b) 110
(c) 115 (d) 120
27. The ratio of milk and water in 66 litre of adulterated milk is 5 : 1. Water is added to it to make the ratio 5 : 3. The quantity of water added is:
(a) 20 litres (b) 22 litres
(c) 24 litres (d) 28 litres
28. A sum of Rs. 41 was divided among 50 children. Each boy gets 90 paise and each girl 65 paise. The number of boys is?
(a) 32 (b) 34
(c) 36 (d) 38
29. Milk and water are mixed in vessel A in the ratio of 5:2 and in vessel B in the ratio of 8 : 5. In what ratio should quantities be taken from the two vessels so as to form a mixture in which milk and water will be in the ratio of 9 : 4?
(a) 7 : 2 (b) 5 : 2
(c) 2 : 7 (d) 2 : 5
30. In a bag, there are three types of coins 1 rupee, 50-paise and 25-paise in the ratio 3 : 8 : 20. Their total value is Rs. 372. The total number of coins is:
(a) 1200 (b) 961
(c) 744 (d) 612

Difficult

1. 8 litres are drawn from a cask full of wine and is then filled with water. This operation is performed three more times. The ratio of the quantity of wine now left in cask to that of water is 16 : 65. How much wine did the cask hold originally?
(a) 18 litres (b) 24 litres
(c) 32 litres (d) 42 litres
2. A can contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and the can is filled with B with same amount the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the can initially?
(a) 10 (b) 20
(c) 21 (d) 25
3. A container contained 80 litre of milk. From this container 8 litre of milk was taken out and replaced by water. This process was further repeated three times. How much of milk is now contained by the container?
(a) 54.23 litre (b) 54.26 litre
(c) 56.26 litre (d) 52.48 litre
4. Vessels A and B contain mixtures of milk and water in the ratios 4 : 5 and 5 : 1 respectively. In what ratio should quantities of mixture be taken from A and B to form a mixture in which milk to water is in the ratio 5:4?

- (a) 2 : 5 (b) 4 : 3
(c) 5 : 2 (d) 2 : 3
5. A can contains a mixture of two liquids A and B in proportion 4 : 7. When 6 litres of mixture are drawn off and the can is filled with B, the proportion of A and B becomes 3 : 7. How many litres of liquids A was contained in the can initially?
(a) 15.4 litres (b) 12.46 litres
(c) 15.7 litres (d) 16.4 litres
6. 100 litres of mixture contains 20% water and the rest is milk. The amount of milk that must be added so that the resulting mixture contains 87.5% milk is:
(a) 30 litres (b) 35 litres
(c) 60 litres (d) 45 litres
7. In what ratio must a person mix three kinds of wheat costing him Rs. 1.20, Rs. 1.44 and Rs. 1.74 per kg so that the mixture may be worth Rs. 1.41 per kg?
(a) 11 : 77 : 7 (b) 25 : 45 : 8
(c) 27 : 23 : 6 (d) None of these
8. Chaiwala mixes three qualities of tea T_1 , T_2 and T_3 priced at Rs. 74 per kg, Rs. 68 per kg, and Rs. 63 per kg in the ratio of 1 : 2 : 4. He added some more quantity of T_1 to 4 kg of this mixture. He sold this new mixture for Rs. 84 per kg, thereby making a profit of 20%. How much of T_1 did he mix with the mixture?
(a) 1 kg (b) 0.5 kg
(c) 0.35 kg (d) 4 kg
9. Nine litres of water are drawn from a container full of water and it is then filled with milk. Nine litres of mixture are drawn and the container is again filled with milk. The quantity of water now left in the container is to that of the milk in it is 16 : 9. How much liquid does the container hold?
(a) 34 litres (b) 38 litres
(c) 43 litres (d) 45 litres
10. A cylinder which has the capacity of eight litres contains a mixture of oxygen and nitrogen. The volume of oxygen in the cylinder is 16% of total volume. A few litres of the mixture is released and an equal amount of nitrogen is added. Then the same amount of the mixture as before is released and replaced by nitrogen for the second time. As a result, the oxygen content becomes 9% of total volume. Find the amount of mixture released each time?
(a) 2 litres (b) 2.5 litres
(c) 1 litre (d) 0.5 litres
11. Two liquids are mixed in the ratio of 3 : 2 and the mixture is sold at Rs. 11 per litre at a 10% profit. If the 1st liquid cost Rs. 2 more per litre than the second, what is the cost per litre of the second liquid?
(a) Rs. 8.80 (b) Rs. 10.80
(c) Rs. 12.80 (d) Rs. 14.80
12. If 5 kg of potato costing Rs. 5 per kg and 3 kg of potato costing Rs. 4 per kg are mixed, find the average cost of the mixture per kg?
(a) Rs. 4.40 (b) Rs. 4.45
(c) Rs. 4.63 (d) Rs. 4.95
13. A 20 litre mixture of milk and water contains milk and water in the ratio 3 : 2. 10 litres of the mixture is removed and replaced with pure milk and the operation is repeated once more. At the end of the two removals and replacement, what is the ratio of milk and water in the resultant mixture?
(a) 17 : 3 (b) 9 : 1
(c) 3 : 17 (d) 5 : 3
14. A merchant mixes three varieties of rice costing Rs. 20 per kg, Rs. 24 per kg and Rs. 30 per kg and sells the mixture at a profit of 20% at Rs. 30 per kg. How many kgs of the second variety will be in the mixture if 2 kgs of the third variety is there in the mixture?
(a) 1 kg (b) 3 kgs
(c) 5 kgs (d) 6 kgs
15. In what ratio must a person mix three kinds of tea costing Rs. 60 per kg, Rs. 75 per kg and Rs. 100 per kg so that the resultant mixture when sold at Rs. 96 per kg yields a profit of 20%?
(a) 1 : 2 : 4 (b) 3 : 7 : 6
(c) 1 : 4 : 2 (d) 1 : 3 : 5
16. 4 kg of potato at Rs. 5 per kg is mixed with 8 kg of potato at Rs. 6 per kg. Find the average price of the mixture?
(a) Rs. 5.66 per kg (b) Rs. 6.50 per kg
(c) Rs. 7.25 per kg (d) None of these
17. A shopkeeper bought 20 kg of onion at Rs. 6.50 per kg and 30 kg of onion at Rs. 7 per kg. He sold the mixture at a profit of Rs. 60. At what price did he sell per kg of the mixture of onions?
(a) Rs. 12 (b) Rs. 8
(c) Rs. 7.50 (d) Rs. 11
18. Two casks of 48 and 42 litres are filled with mixtures of wine and water, the proportions in the two casks being respectively 13 : 7 and 18 : 17. If the contents of the two casks be mixed, and 20 litres of water added to the whole what will be the proportion of wine to water in the resultant mixture?
(a) 13 : 12 (b) 12 : 13
(c) 21 : 31 (d) 31 : 21
19. Three glasses of capacity 2 litres, 5 litres and 9 litres contain mixture of milk and water with milk concentrations 90%, 80% and 70% respectively. The contents of three glasses are emptied into a large vessel. Find the ratio of milk to water in the resultant mixture?
(a) 121 : 39 (b) 131 : 49
(c) 39 : 121 (d) 49 : 131

20. A 12-litres cylinder contains a mixture of oxygen and nitrogen, the volume of oxygen being 40% of total volume. A few litres of the mixture is released and an equal amount of nitrogen is added. Then the same amount of the mixture as before is released and

replaced by nitrogen for the second time. As a result, the oxygen content becomes 10% of the total volume. How many litres of mixture is released each time?

- (a) 3 litres (b) 9 litres
(c) 6 litres (d) 4 litres

Previous Year (Memory Based)

- In an alloy there is 12% of copper. To get 69 kg of copper. How much alloy will be required?
(a) 424 kg (b) 575 kg
(c) 828 kg (d) $1736 \frac{2}{3}$ kg
- 40 litres of a mixture of milk and water contains 10% of water, the water to be added, to make the water content 20% in the new mixture is:
(a) 6 litres (b) 6.5 litres
(c) 5.5 litres (d) 5 litres
- How much pure alcohol has to be added to 400 ml of a solution containing 15% of alcohol to change the concentration of alcohol in the mixture to 32%?
(a) 60 ml (b) 100 ml
(c) 128 ml (d) 68 ml
- In what ratio must a grocer mix tea at Rs. 60 a kg. and Rs. 65 a kg, so that by selling the mixture at Rs. 68.20 a kg, he may gain 10%.
(a) 3 : 2 (b) 3 : 4
(c) 3 : 5 (d) 4 : 5
- In 50 gm alloy of gold and silver. The gold is 80% by weight. How much gold should be mixed to this alloy so that the weight of gold would become 95%?
(a) 200 gm (b) 150 gm
(c) 50 gm (d) 10 gm
- A person has a milk of Rs. 50 per litre. In what ratio should water be mixed with milk so that on selling the mixture at Rs. 40 litres he may get profit of 25%.
(a) 14 : 9 (b) 16 : 9
(c) 9 : 14 (d) 9 : 16
- 7 kg of tea costing Rs. 280 per kg is mixed with 9 kg of tea costing Rs. 240 per kg. The average price per kg of the mixed tea is:
(a) Rs. 255.80 (b) Rs. 257.50
(c) Rs. 267.20 (d) Rs. 267.50
- In what ratio must a mixture of 30% alcohol strength be mixed with that of 50% alcohol strength so as to get a mixture of 45% alcohol strength:
(a) 1 : 2 (b) 1 : 3
(c) 2 : 1 (d) 3 : 1
- In a 40 litres mixture of milk and water the ratio of milk to water is 7 : 1. In order to make the ratio of milk and water 3 : 1, the quantity of water (in litres) that should be added to the mixture will be:
- A jar contained a mixture of two liquids A and B in the ratio 7 : 2. When 18 litres of mixture was taken out and 18 litres of liquid B was poured into the jar. This ratio became 2 : 3. The quantity of liquid A contained in the jar initially was:
(a) $\frac{450}{17}$ litres (b) $\frac{490}{19}$ litres
(c) $\frac{490}{17}$ litres (d) $\frac{450}{19}$ litres
- In a mixture of 30 litres, the ratio of milk to water 2 : 1. The amount of water to be further added to the mixture so as to make the ratio of the milk to water 1 : 2 will be
(a) 45 litres (b) 60 litres
(c) 30 litres (d) 40 litres
- A and B are two alloys of gold and copper prepared by mixing metals in the ratio 5 : 3 and 5 : 11 respectively. Equal quantities of these alloys are melted to form a third alloy C. The ratio of gold and copper in the alloy C is:
(a) 25 : 13 (b) 33 : 15
(c) 15 : 17 (d) 17 : 15
- An alloy contains zinc, copper and tin in the ratio 2 : 3 : 1 and another contains copper, tin and lead in the ratio 5 : 4 : 3. If equal weights of both alloys are melted together to form a third alloy, then the weight of lead per kg in the new alloy will be
(a) $\frac{1}{2}$ kg (b) $\frac{1}{8}$ kg
(c) $\frac{3}{14}$ kg (d) $\frac{7}{9}$ kg
- The ratio in which tea costing Rs. 192 per kg is to be mixed with tea costing Rs. 150 per kg so that the mixed tea, when sold for Rs. 194.40 per kg, gives a profit of 20% is
(a) 2 : 5 (b) 23 : 5
(c) 5 : 3 (d) 5 : 2

15. In one glass, milk and water are mixed in the ratio 3 : 5 and in another glass they are mixed in the ratio 6 : 1. In what ratio should the content of the two glasses be mixed together so that the new mixture contains milk and water in the ratio 1 : 1?
 (a) 20 : 7 (b) 8 : 3
 (c) 27 : 4 (d) 25 : 9
16. In what ratio Darjeeling Tea costing Rs. 320 per kg be mixed with Assam Tea costing Rs. 250 per kg so that there is a gain of 20% by selling the mixture at Rs. 324 per kg?
 (a) 1 : 2 (b) 2 : 3
 (c) 3 : 2 (d) 2 : 5
17. The ratio of the volumes of water and glycerine in 240cc of mixture is 1 : 3. The quantity of water (in cc) that should be added to the mixture so the volumes of water and glycerine becomes 2 : 3 is:
 (a) 55 (b) 60
 (c) 62.5 (d) 64
18. Two vessels A and B contains acid and water in the ratio 4 : 3 and 5 : 3 respectively. Then the ratio in which these mixture to be mixed to obtain a new mixture in vessel C containing acid and water in the ratio 3 : 2 is
 (a) 5 : 8 (b) 7 : 8
 (c) 7 : 5 (d) 4 : 7
19. Acid and water are mixed in a vessel A in the ratio of 5 : 2 and in the vessel B in the ratio 8 : 5. In what proportion should quantities be taken out from the two vessels so as to form a mixture in which the acid and water will be in the ratio of 9 : 4?
 (a) 7 : 2 (b) 2 : 7
 (c) 7 : 4 (d) 2 : 3
20. The ratio of spirit and water in two mixtures of 20 liters and 36 litres is 3 : 7 and 7 : 5 respectively. Both the mixtures are mixed together. Now the ratio of the spirit and water in the new mixture is
 (a) 25 : 29 (b) 9 : 10
 (c) 27 : 29 (d) 27 : 31
21. In two types of stainless steel, the ratio of chromium and steel are 2 : 11 and 5 : 21 respectively. In what proportion should the two types be mixed so that the ratio of chromium to steel in the mixed type becomes 7 : 32?
 (a) 2 : 3 (b) 3 : 4
 (c) 1 : 2 (d) 1 : 3
22. A and B are two alloys of gold and copper in the ratio 7 : 2 and 7 : 11 respectively. If equal quantities of these two alloys are melted to form a new alloy C, then the ratio of gold and copper in C is:
 (a) 6 : 5 (b) 9 : 4
 (c) 12 : 7 (d) 7 : 5
23. A can contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drained off and the can is filled with B, the ratio of A and B becomes. 7 : 9. How many litres of liquid A was contained by the can initially?
 (a) 10 (b) 20
 (c) 21 (d) 35
24. A shopkeeper bought 15 kg of rice at the rate of Rs. 29 per kg and 25 kg of rice at the rate of Rs. 20 per kg. He sold the mixture of both types of rice at the rate of Rs. 27 per kg. His profit in this transaction is:
 (a) Rs. 125 (b) Rs. 150
 (c) Rs. 140 (d) Rs. 145
25. The proportion of acid and water in three samples is 2 : 1, 3 : 2 and 5 : 3. A mixture containing equal quantities of all three samples is made. The ratio of water and acid in the mixture is
 (a) 120 : 133 (b) 227 : 133
 (c) 227 : 120 (d) 133 : 227
26. Nikita bought 20 kg of wheat at the rate of Rs. 9.50 per kg and 30 kg of wheat at the rate of Rs. 8.50 per kg and mixed them. She sold the mixture at the rate of Rs. 8.90 per kg. Her total profit or loss in the transaction was:
 (a) Rs. 2 loss (b) Rs. 2 profit
 (c) Rs. 7 loss (d) No profit no loss
27. A mixture of a certain quantity of milk with 16 litres of water is worth Rs. 1.5 per litre. If pure milk be worth Rs. 3 per litre. How much milk is there in the mixture?
 (a) 40 litres (b) 16 litres
 (c) 60 litres (d) 80 litres
28. Three vessels whose capacities are 3 : 2 : 1 are completely filled with milk mixed with water. The ratio of milk and water in the mixture of vessels are 5 : 2, 4 : 1 and 4 : 1 respectively. Taking $\frac{1}{3}$ of first, $\frac{1}{2}$ of second and $\frac{1}{7}$ of third mixtures, a new mixtures, kept in a new vessel is prepared. The percentage of water in the new mixture is:
 (a) 32 (b) 28
 (c) 30 (d) 24
29. In 2 kg mixture of copper and aluminium, 30% is copper. How much aluminium powder should be added to the mixture so that the quantity of copper becomes 20% ?
 (a) 900 gms (b) 800 gms
 (c) 1000 gms (d) 1200 gms
30. The liquids X and Y are mixed in the ratio of 3 : 2 and the mixture is sold at Rs. 11 per litre at a profit of 10%. If the liquid X costs Rs. 2 more per litre than Y, the cost of X per litre is (in Rs.):
 (a) 9.50 (b) 10.80
 (c) 11.75 (d) 11

Foundation

Solutions

1. (b); In the mixture quantity of milk are $\frac{7}{12} \times 60 = 35$ L

In mixture quantity of water are $\frac{5}{12} \times 60 = 25$ litre

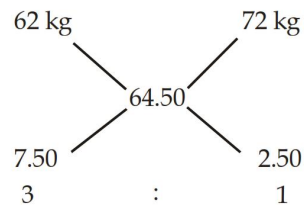
Quantity of water is added by $= 35 - 25 = 10$ L.

2. (a); Quantity of milk & water be $\left(\frac{30}{15} \times 8\right), \left(\frac{30}{15} \times 7\right)$

Milk = 16 Litre, Water = 14 Litre
Let x litre water be added

$$\frac{16}{14+x} = \frac{4}{5} \Rightarrow 56 + 4x = 80 \Rightarrow x = 6$$

3. (c); By mixture Alligation method:



4. (b); Quantity of water in a mixture of 60 litre is

$$= \frac{20}{100} \times 60 = 12 \text{ litre}$$

quantity of milk = 60 lt. - 12 lt. = 48 lt.

In new mixture Water = 25%, Milk = 75%

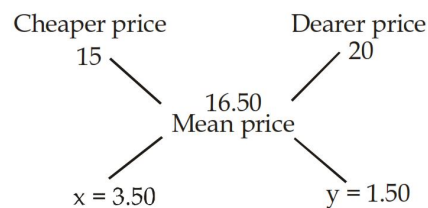
Let total quantity of new mixture be x litre

$$x \times \frac{75}{100} = 48 \Rightarrow x = 64$$

So, quantity of water = 64 - 48 = 16 litre

So, 4 ltr. water must be added

5. (c); By the rule of alligation:

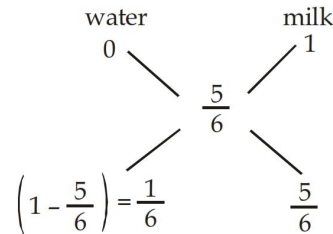


$\therefore$ Required Ratio = 3.50 : 1.50 = 70 : 30 = 7 : 3

6. (b); Let CP of milk be 1 per litre.
Then SP of 1 litre of mixture = 1 Rs.
Gain obtained = 20%

$$\text{CP of 1 litre of mixture} = \frac{100}{120} \times 1 = \frac{5}{6}$$

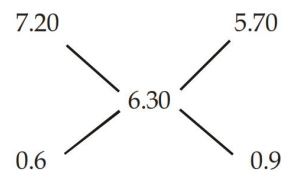
By the rule of alligation, we have



$\therefore$ Ratio of water and milk = $\frac{1}{6} : \frac{5}{6} = 1 : 5$

$$\text{Required percentage} = \frac{1}{6} \times 100\% = 16.66\%$$

7. (b); By the rule of alligation

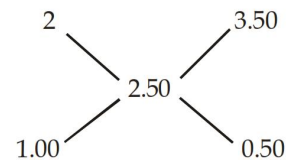


Required Ratio = 6 : 9 = 2 : 3

8. (b); The required ratio will be same as the ratio of amount of their investment.

$\therefore$ Required Ratio = 45000 : 30000 = 3 : 2

9. (a); By the rule of alligation



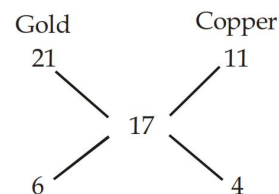
$\therefore$ Required ratio = 1 : 1 = 1 : 1

10. (c); Quantity of sugar in solution = 240 gram

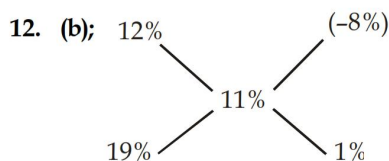
Let quantity added = x gram

$$\therefore \frac{240+x}{360} = \frac{1}{1} \Rightarrow x = 120 \text{ grams}$$

11. (d); By the rule of alligation



Required ratio = 6 : 4 = 3 : 2



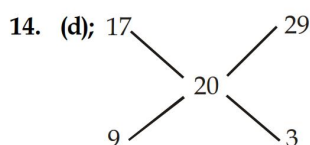
$$\frac{\text{books sold at 12\% profit}}{\text{books sold at 8\% loss}} = \frac{19}{1}$$

$$\therefore \text{Required quantity of Book} = \frac{19}{20} \times 100 = 19 \times 5 = 95$$

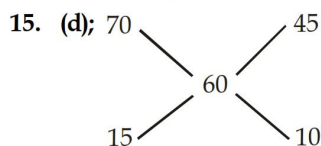
13. (b); Milk = $\frac{5}{7} + \frac{4}{7} + \frac{3}{4} = \frac{20+16+21}{28} = \frac{57}{28}$

$$\text{Water} = \frac{2}{7} + \frac{3}{7} + \frac{1}{4} = \frac{8+12+7}{28} = \frac{27}{28}$$

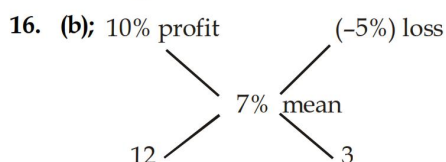
$$\text{milk : water} = \frac{57}{28} : \frac{27}{28} = 57 : 27 = 19 : 9$$



$$\therefore \text{Required proportion} = 9 : 3 = 3 : 1$$

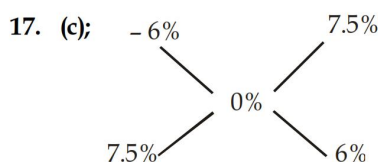


$$\text{Required ratio} = 15 : 10 = 3 : 2$$



$$\frac{\text{quantity of rice sells at 10\% profit}}{\text{quantity of rice sells at 5\% loss}} = \frac{12}{3} = \frac{4}{1}$$

$$\therefore \text{Required quantity} = \frac{4}{5} \times 50 = 40 \text{ kg.}$$



$$\therefore \text{Ratio} = 7.5 : 6 = 75 : 60 = 5 : 4$$

$$\therefore \text{2nd cow cost} = \frac{4}{9} \times 2700 = \text{Rs. 1200}$$

18. (d); Let Nutan invested x Rs.

$$\therefore \text{Sonal's contribution} = 5000 + x$$

$$\therefore \text{Aditya's contribution} = 9000 + x$$

$$\therefore x + (5000 + x) + (9000 + x) = 50000$$

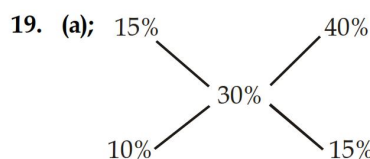
$$3x + 14000 = 50000$$

$$3x = 36000 \Rightarrow x = 12000$$

$$\therefore \text{Ratio} = 21000 : 17000 : 12000 = 21 : 17 : 12$$

$$\text{Amount that Aditya receive} = \frac{21}{50} \times 70000$$

$$= 21 \times 1400 = 29400$$



$$\therefore \text{Required ratio} = 10 : 15 = 2 : 3$$

20. (a); CP of 30 kgs = $30 \times 9.50 = 285$

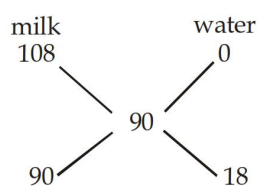
$$\text{CP of 40 kgs} = 40 \times 8.50 = 340$$

$$\text{Total CP of 70 kgs} = 285 + 340 = 625$$

$$\text{SP of 70 kgs} = 70 \times 8.90 = 623$$

$$\therefore \text{Loss} = 625 - 623 = \text{Rs. 2}$$

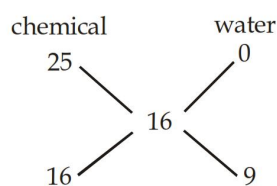
21. (d); The mean value is 90 paisa and the price of water is zero paisa



$$\text{Ratio of milk and water} = 5 : 1$$

$$\text{Quantity of milk in the mixture} = 16 \times 5 = 80 \text{ ltr.}$$

22. (b); CP of mixture = $\frac{100}{(100+25)} \times 20 = \frac{100}{125} \times 20 = 16$



$$\therefore \text{Required ratio} = 16 : 9$$

23. (c); Quantity of milk in the mixture

$$= \frac{90}{100} \times 40 = 36 \text{ ltr.}$$

Quantity of water in the mixture

$$= \frac{10}{100} \times 40 = 4 \text{ ltr.}$$

Let x ltr. of water be added

$$\frac{36}{4+x} = \frac{80}{20} \Rightarrow (36) = 4(4+x)$$

$$36 = 16 + 4x \Rightarrow 4x = 20 \Rightarrow x = 5 \text{ Ltr.}$$

24. (b); Quantity of zinc in metal 1st = $\frac{1}{3} \times 2 = \frac{2}{3}$

Quantity of copper in metal 1st = $\frac{2}{3} \times 2 = \frac{4}{3}$

Quantity of zinc in mixture = $\frac{2}{3} + \frac{3}{4} = \frac{8+9}{12} = \frac{17}{12}$

Quantity of copper in mixture = $\frac{2}{3} \times 2 + \frac{3}{4} \times 3$

$$= \frac{4}{3} + \frac{9}{4} = \frac{16+27}{12} = \frac{43}{12}$$

$\therefore$ Required ratio = $\frac{17}{12} : \frac{43}{12} = 17 : 43$

25. (b); Let quantity of gold to be mixed = x g

$$\frac{\frac{80}{100} \times 50 + x}{\frac{20}{100} \times 50} = \frac{95}{5} \Rightarrow \frac{40+x}{10} = \frac{19}{1}$$

$$40 + x = 190 \Rightarrow x = 150 \text{ g}$$

26. (c); Let, 5-paise coin = x, 10-paise coin = y
 $x + y = 80$... (i)

$$\frac{5}{100} \times x + \frac{10}{100} \times y = 6.40 \Rightarrow \frac{x}{20} + \frac{y}{10} = 6.40$$

$$\frac{x+2y}{20} = 6.40 \Rightarrow x + 2y = 128$$
 ... (ii)

From (i) and (ii)

$$x + y - x - 2y = 80 - 128$$

$$-y = -48 \Rightarrow y = 48 \Rightarrow x = 80 - 48 = 32$$

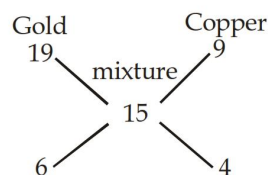
27. (d); By the rule of alligation



$\therefore$ Ratio of tea = 7 : 5

$\therefore$ Required quantity = $7 \times 3 = 21 \text{ kg.}$

28. (b); By the rule of alligation



Required ratio = 6 : 4 = 3 : 2

29. (d); By alligation formula:

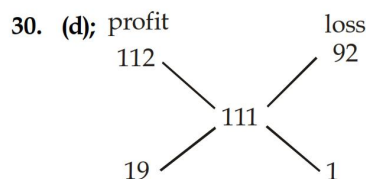
Quantity of milk left after 2 operations

$$= x \left(1 - \frac{y}{x} \right)^2 = 81 \left(1 - \frac{27}{81} \right)^2 = 81 \times \frac{2}{3} \times \frac{2}{3}$$

$$= 4 \times 9 = 36 \text{ Ltr.}$$

and water quantity in mixture = $81 - 36 = 45 \text{ Ltr.}$

required ratio = 36 : 45 = 4 : 5



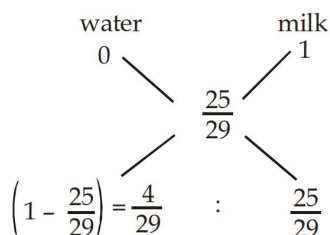
no. of pens were sold at 12% profit = $\frac{19}{20} \times 60 = 57$

Moderate

1. (b); Let cost price of 1 litre of milk = Rs. 1

SP of 1 litre of mixture = Rs. 1; Gain = 16%

CP of 1 litre of mixture = $\left(\frac{100}{100+16} \right) \times 1 = \frac{25}{29}$

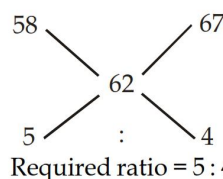


Required ratio = $\frac{4}{29} : \frac{25}{29} = 4 : 25$

Shortcut:

Ratio = Profit% : 100 = 16 : 100 = 4 : 25

2. (d); CP of mixture = $\frac{100}{110} \times 68.20 = 62$



Required ratio = 5 : 4

3. (b); Let cost of 1 litre of milk be Rs. 1

$$\text{Milk in 1 litre mix in 1st can} = \frac{3}{4}$$

$$\text{Milk in 1 litre mix in 2nd can} = \frac{1}{2}$$

$$\text{Milk in 1 litre of mix solution} = \frac{5}{8}$$

$$\text{Mean price} = \frac{5}{8}$$

$$\begin{array}{ccc} \frac{3}{4} & & \frac{1}{2} \\ & \searrow \quad \nearrow & \\ & \frac{5}{8} & \\ & \nearrow \quad \searrow & \\ \frac{1}{8} & & \frac{1}{8} \end{array}$$

$$\text{Ratio} = \frac{1}{8} : \frac{1}{8} = 1 : 1$$

$$\text{quantity of milk from each can} = \frac{1}{2} \times 12 = 6 \text{ litre}$$

4. (d); CP of mixture = $\frac{100}{110} \times 9.24 = \frac{42}{5} = 8.4$

(By rule of alligation)

$$\begin{array}{ccc} 9 & & 7 \\ & \searrow \quad \nearrow & \\ & 8.4 & \\ & \nearrow \quad \searrow & \\ 1.4 & & 0.6 \end{array}$$

$$\therefore \text{Ratio} = 14 : 6 = 7 : 3$$

$$\text{Quantity to be added} = 9 \times 7 = 63 \text{ kg.}$$

5. (d); By rule:

$$\text{Quantity of milk} = x \left(1 - \frac{y}{x} \right)^n$$

$$= 50 \left(1 - \frac{5}{40} \right)^3 = 50 \left(\frac{9}{10} \right)^3 = \frac{50 \times 9 \times 9 \times 9}{10 \times 10 \times 10}$$

$$= 36.45 \text{ litres}$$

6. (d); Let quantity to be added = x litre

$$\therefore \frac{9+x}{21} = \frac{2}{1} \Rightarrow 9+x = 42 \Rightarrow x = 42 - 9$$

$$x = 33 \text{ litre}$$

7. (a); Let the price per kg of the mixture = x

$\therefore$ By the rule of Alligation

$$\begin{array}{ccc} 15 & & 20 \\ & \searrow \quad \nearrow & \\ & x & \\ & \nearrow \quad \searrow & \\ 2 & & 3 \end{array}$$

$$\therefore \frac{x-15}{20-x} = \frac{3}{2} \Rightarrow 2x - 30 = 60 - 3x$$

$$5x = 90 \Rightarrow x = 18$$

8. (c); By the rule of alligation

$$\begin{array}{ccc} 8\% & & 18\% \\ & \searrow \quad \nearrow & \\ & 14\% & \\ & \nearrow \quad \searrow & \\ 4\% & & 6\% \end{array}$$

$$\therefore \text{ratio} = 4 : 6 = 2 : 3$$

$$\text{recquired quantity} = \frac{3}{5} \times 1000 = 600 \text{ kg}$$

9. (c); By the rule of alligation

$$\begin{array}{ccc} 85\% & & 70\% \\ & \searrow \quad \nearrow & \\ & 75\% & \\ & \nearrow \quad \searrow & \\ 5\% & & 10\% \end{array}$$

$$\therefore \text{ratio of boys and girls} = 1 : 2$$

$$\therefore \text{no. of girls passed} = \frac{2}{3} \times 900 = 2 \times 300 = 600$$

10. (d); Quantity of salt in 400 gm of salt solution

$$= \frac{40}{100} \times 400 = 160 \text{ gm}$$

$$\text{Let quantity added} = x$$

$$\therefore \frac{160+x}{(400-160)} = \frac{1}{1} \Rightarrow 160 + x = 240 \Rightarrow x = 80$$

11. (d); Let price per dozen of the mixture = x

$\therefore$ By the rule of alligation

$$\begin{array}{ccc} 12 & & 10 \\ & \searrow \quad \nearrow & \\ & x & \\ & \nearrow \quad \searrow & \\ 3 & & 5 \end{array}$$

$$\therefore \frac{12-x}{x-10} = \frac{5}{3} \Rightarrow 36 - 3x = 5x - 50 \Rightarrow 8x = 86$$

$$x = 10.75$$

12. (a); Let mean price = x

$$\begin{array}{ccc} 4.20 & & 5.40 \\ & \searrow \quad \nearrow & \\ & x & \\ & \nearrow \quad \searrow & \\ 5.40 - x & & x - 4.20 \end{array}$$

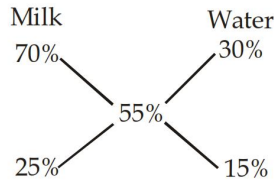
$$\frac{x-4.2}{5.4-x} = \frac{5}{3} \Rightarrow 8x = 39.60 \Rightarrow x = 4.95$$

13. (b); Let quantity to be added = x

$$\therefore \frac{43.2}{4.8+x} = \frac{80}{20}$$

$$86.4 = 38.4 + 8x \Rightarrow 8x = 48 \Rightarrow x = 6$$

14. (b); By the rule of alligation



$$\text{ratio} = 25 : 15 = 5 : 3$$

$$\therefore \text{Quantity of milk} = \frac{5}{8} \times 160 = 100 \text{ ltr.}$$

$$\text{and, Quantity of water} = \frac{3}{8} \times 160 = 60 \text{ ltr.}$$

15. (c); Let amount of water to be added = x

$$\therefore \frac{72}{8+x} = \frac{80}{20} \Rightarrow 72 = 32 + 4x$$

$$4x = 40 \Rightarrow x = 10$$

16. (d); Let initially the amount of liquid P in the tub = 4x
initially the amount of liquid Q in the tub = x

$$\therefore \frac{4x - \frac{4}{5} \times 10}{x - \frac{1}{5} \times 10 + 10} = \frac{2}{3} \Rightarrow \frac{4x - 8}{x - 2 + 10} = \frac{2}{3}$$

$$12x - 24 = 2x + 16 \Rightarrow 10x = 40 \Rightarrow x = 4$$

$$\therefore \text{Quantity of liquid P} = 4 \times 4 = 16 \text{ litre}$$

17. (d); Quantity of milk in new mixture-

$$= x \left(1 - \frac{y}{x}\right)^x = 27 \left(1 - \frac{9}{27}\right)^2 = 27 \times \frac{2}{3} \times \frac{2}{3} = 12 \text{ litre.}$$

$$\text{Quantity of water in new mixture} = 27 - 12 = 15 \text{ litre}$$

$$\text{Required Ratio} = 12 : 15 = 4 : 5$$

18. (d); Share in profit = $\frac{495}{1020 - 495} = \frac{495}{525} = \frac{33}{35}$

Let Aditya invested his money for x month then
Manish will invest (36 - x).

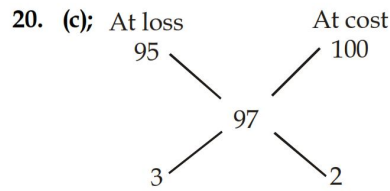
$$\therefore \frac{300 \times x}{500(36-x)} = \frac{33}{35} \Rightarrow \frac{x}{36-x} = \frac{11}{7} \Rightarrow x = 22$$

19. (c); Let Initial quantity of A and B = 4x and x

$$\therefore \frac{3x - \frac{3}{5} \times 5}{2x - \frac{2}{5} \times 5 + 5} = \frac{2}{3} \Rightarrow \frac{3x-3}{2x+3} = \frac{2}{3}$$

$$9x - 9 = 4x + 6 \Rightarrow 5x = 15 \Rightarrow x = 3$$

$$\therefore \text{quantity of A initially} = 3 \times 3 = 9 \text{ litres.}$$



$$\therefore \text{Ratio} = \frac{3}{2}$$

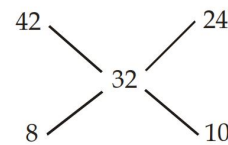
$$\text{Quantity sold at cost price} = \frac{2}{5} \times 40 = 16 \text{ kg.}$$

21. (c); Let amount of water to be added = x

$$\therefore \frac{50}{25+x} = \frac{1}{2} \Rightarrow 100 = 25 + x \Rightarrow x = 75 \text{ ltr.}$$

22. (a); CP of the mixture = $\frac{100}{125} \times 40 = \frac{4}{5} \times 40 = 32$

By the rule of alligation



$$\text{ratio of salts} = 4 : 5$$

$$\therefore \text{required quantity} = 5 \times 4 = 20$$

23. (c); Let sugar to be added = x

$$\therefore \frac{120+x}{180} = \frac{1}{1} \Rightarrow x = 60$$

24. (d); When mix 40 litres of water with 100 litres of milk.

$$\therefore \text{Required proportion} = 40 : 100 = 2 : 5$$

$$\therefore \% \text{ water} = \frac{2}{2+5} \times 100 = 28.56\%$$

25. (c); Water in 200 litre mixture = $\frac{15}{100} \times 200 = 30 \text{ L}$

$$\text{milk in 200 litre mixture} = (200 - 30) = 170 \text{ litre}$$

$$170 + x = \frac{875}{100 \times 10} \times (200 + x)$$

$$170 + x = \frac{7}{8}(200 + x) \Rightarrow 1360 + 8x = 1400 + 7x$$

$$x = 40 \text{ litre}$$

26. (b); Let no. of rabbits = x, no. of pigeons = y

$$\therefore x + y = 200 \quad \dots (i)$$

$$4x + 2y = 580 \quad \dots (ii)$$

equation (i) $\times$ (ii) - equation (ii)

$$2x + 2y = 400$$

$$\underline{4x + 2y = 580}$$

$$\underline{-2x = -180}$$

$$x = 90 \Rightarrow y = 110.$$

27. (b); Let x be the quantity to added; $\frac{55}{11+x} = \frac{5}{3}$
 $165 = 55 + 5x \Rightarrow 5x = 110 \Rightarrow x = 22$

28. (b); average sum = $\frac{4100}{50} = 82$ paisa

Let no. of boy = x , no. of girls = y

By the rule of alligation

$$\begin{array}{ccc} 90 & & 65 \\ & \searrow \quad \swarrow & \\ & 82 & \\ & \swarrow \quad \searrow & \\ x & & y \end{array}$$

$$\therefore \frac{x}{y} = \frac{17}{8} \Rightarrow \text{no. of boys} = \frac{17}{25} \times 50 = 34$$

29. (a);

$$\begin{array}{ccc} A & & B \\ \frac{5}{7} & & \frac{8}{13} \\ & \searrow \quad \swarrow & \\ & \frac{9}{13} & \\ & \swarrow \quad \searrow & \\ \frac{1}{13} & & \frac{2}{91} \left(\frac{5}{7} - \frac{9}{13} \right) \end{array}$$

$$\text{Required Ratio} = \frac{1}{13} : \frac{2}{91} = 7 : 2$$

30. (b); Let no. of coins of 1 rupee = $3x$
 no. of coins of 50 paisa = $8x$
 no. of coins of 25 paisa = $20x$

$$\therefore 3x + \frac{8x}{2} + \frac{20x}{4} = 372 \Rightarrow 3x + 4x + 5x = 372$$

$$12x = 372 \Rightarrow x = 31$$

$$\therefore \text{Total no. of coins} = 31 \times 31 = 961$$

Difficult

1. (b); Let original quantity = x Ltr.
 Amount of wine left after 4 operations

$$\frac{16}{81} = \frac{x \left(1 - \frac{8}{x}\right)^4}{x} \Rightarrow \left(1 - \frac{8}{x}\right)^4 = \left(\frac{2}{3}\right)^4$$

$$1 - \frac{8}{x} = \frac{2}{3} \Rightarrow \frac{8}{x} = 1 - \frac{2}{3} = \frac{1}{3} \Rightarrow x = 24 \text{ Ltr.}$$

2. (c); Let us suppose that the Can initially contains $7x$ and $5x$ litres of mixture A and B respectively

$$\therefore \text{Quantity of A in mixture left} = \left(7x - \frac{7}{12} \times 9\right)$$

$$= 7x - \frac{21}{4} \text{ Ltr.}$$

$$\text{Quantity of B in mixture left} = \left(5x - \frac{5}{12} \times 9\right)$$

$$= 5x - \frac{15}{4} \text{ Ltr.}$$

$$\therefore \frac{7x - \frac{21}{4}}{\left(5x - \frac{15}{4}\right) + 9} = \frac{7}{9} \Rightarrow \frac{28x - 21}{20x + 21} = \frac{7}{9}$$

$$252x - 189 = 140x + 147 \Rightarrow 112x = 336 \Rightarrow x = 3$$

$$\therefore \text{quantity contained by A initially} = 7x = 21 \text{ Ltr.}$$

3. (d); Milk contained in container = $80 \left(1 - \frac{8}{80}\right)^4$

$$= 80 \times \left(\frac{72}{80}\right)^4 = 80 \times \left(\frac{9}{10}\right)^4 = 80 \times (0.9)^4$$

$$= 80 \times 0.6561 = 52.48 \text{ Ltr.}$$

4. (c); Quantity of milk in vessel A = $\frac{4}{9}$

$$\text{Quantity of milk in vessel B} = \frac{5}{6}$$

$$\text{Quantity of milk in mixture} = \frac{5}{9}$$

$$\begin{array}{ccc} \frac{4}{9} & & \frac{5}{6} \\ & \searrow \quad \swarrow & \\ & \frac{5}{9} & \\ & \swarrow \quad \searrow & \\ \frac{5}{18} & & \frac{1}{9} \end{array}$$

$$\therefore \text{required ratio} = \frac{5}{18} : \frac{9}{1} = 5 : 2$$

5. (b); According to question

$$\frac{4x - \frac{4}{11} \times 6}{7x - \frac{7}{11} \times 6 + 6} = \frac{3}{7} \Rightarrow x = \frac{240}{77}$$

$$4x = \frac{960}{77} = 12.46$$

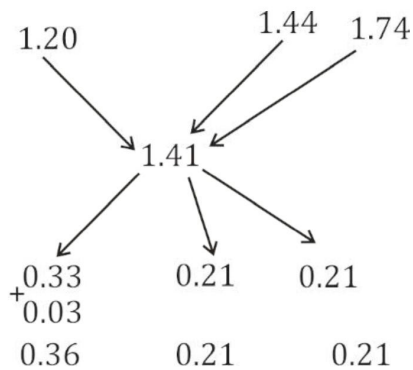
6. (c); Milk in 100 ltr. of mixture = $\frac{80}{100} \times 100 = 80$ ltr.

Water quantity = 20 ltr.

$$\therefore \frac{80 + x}{20} = \frac{87.5}{12.5}$$

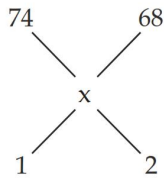
$$\frac{80 + x}{20} = 7 \Rightarrow 80 + x = 140 \Rightarrow x = 60 \text{ ltr.}$$

7. (d); Using by allegations



$\therefore$ Required ratio = 12 : 7 : 7

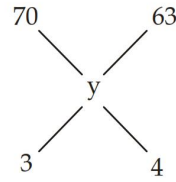
8. (d); Let first take T_1 and T_2



$$\therefore \frac{74 - x}{x - 68} = \frac{2}{1} \Rightarrow 74 - x = 2x - 136$$

$$3x = 210 \Rightarrow x = 70$$

Now take T_3

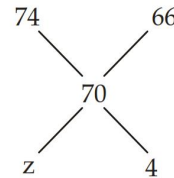


$$\frac{70 - y}{y - 63} = \frac{4}{3} \Rightarrow 210 - 3y = 4y - 252$$

$$7y = 462 \Rightarrow y = 66$$

Now Let us assume that z quantity of T_1 is mixed with mixture

$$CP = 84 \times \frac{100}{120} = 70$$



$$\frac{74 - 70}{70 - 66} = \frac{4}{z} \Rightarrow \frac{4}{4} = \frac{4}{z} \Rightarrow z = 4 \text{ kg.}$$

9. (d); Let initially container contain x litre of water

$$\therefore \frac{x \left(1 - \frac{9}{x}\right)^2}{x} = \frac{16}{25} \Rightarrow \left(1 - \frac{9}{x}\right)^2 = \left(\frac{4}{5}\right)^2$$

$$1 - \frac{9}{x} = \frac{4}{5} \Rightarrow \frac{9}{x} = 1 - \frac{4}{5} = \frac{1}{5} \Rightarrow x = 45 \text{ litres}$$

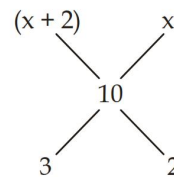
10. (a); Let each time y amount of mixture releases

$$\therefore \frac{8 \left(1 - \frac{y}{8}\right)^2}{8} = \frac{0.09}{0.16} \Rightarrow \left(1 - \frac{y}{8}\right)^2 = \left(\frac{3}{4}\right)^2$$

$$\frac{y}{8} = \frac{1}{4} \Rightarrow y = 2 \text{ litres.}$$

11. (a); Let second liquid costs = x per litre
Let first liquid costs = (x + 2) per litre

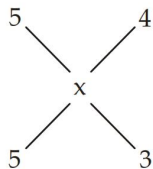
$$CP = 11 \times \frac{100}{110} = \text{Rs. } 10$$



$$\therefore \frac{x+2-10}{10-x} = \frac{2}{3} \Rightarrow \frac{x-8}{10-x} = \frac{2}{3}$$

$$3x - 24 = 20 - 2x \Rightarrow 5x = 44 \Rightarrow x = 8.8 \text{ per litre.}$$

12. (c); Let average cost = x per kg.



$$\frac{5-x}{x-4} = \frac{3}{5} \Rightarrow 25 - 5x = 3x - 12$$

$$8x = 37 \Rightarrow x = 4.625 = 4.63 \text{ (Approximate)}$$

13. (b); The 20 litre mixture contains milk and water in the ratio of 3 : 2. Therefore, there will be 12 litres of milk in the mixture and 8 litres of water in the mixture

Step 1: When 10 litres of the mixture is removed, 6 litres of milk is removed and 4 litres of water is removed. Therefore, there will be 6 litres of milk and 4 litres of water left in the container. It is then replaced with pure milk of 10 litres. Now the container will have 16 litres of milk and 4 litres of water.

Step 2: When 10 litres of the new mixture is removed, 8 litres of milk and 2 litres of water is removed. The container will have 8 litres of milk and 2 litres of water in it. Now 10 litres of pure milk is added. Therefore, the container will have 18 litres of milk and 2 litres of water in it at the end of the second step.

So, required ratio = 18 : 2 or 9 : 1.

14. (c); If the selling price of mixture is Rs. 30/kg and the merchant makes a profit of 20%, then the cost

$$\text{price of the mixture} = \frac{30}{1.2} = \text{Rs. } 25 / \text{kg.}$$

We need to find out the ratio in which the three varieties are mixed to obtain a mixture costing Rs. 25/kg.

Let Variety A costs Rs. 20/kg, Variety B costs Rs. 24/kg and variety C costs Rs. 30/kg. The mean desired price falls between B and C.

Find out the ratio $Q_A : Q_C$ using alligation rule

$$\frac{Q_A}{Q_C} = \frac{30-25}{25-20} = \frac{5}{5} = 1$$

Find out the ratio $Q_B : Q_C$ using alligation rule

$$\frac{Q_B}{Q_C} = \frac{30-25}{25-24} = \frac{5}{1} = 5$$

Q_C is found by adding the value of Q_C in the

above two cases, i.e; $Q_C = 1 + 1 = 2$

Therefore, the required ratio = 1 : 5 : 2

If there are 2 kgs of the third variety in the mixture, then there will be 5 kgs. of the second variety in the mixture.

15. (c); The resultant mixture is sold at a profit of 20% at Rs. 96/kg.

$$\Rightarrow \text{Cost price} = \frac{96}{1.2} = \text{Rs. } 80 / \text{kg.}$$

Let the three varieties be A, B and C costing Rs. 60, Rs. 75 and Rs. 100 respectively.

The mean price falls between B and C

Hence the following method should be used:

Step 1: Find out the ratio of $Q_A : Q_C$ using alligation rule

$$\frac{Q_A}{Q_C} = \frac{100-80}{80-60} = \frac{20}{20} = 1$$

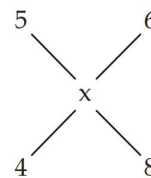
Step 2: Find out the ratio of $Q_B : Q_C$ using alligation rule

$$\frac{Q_B}{Q_C} = \frac{100-80}{80-75} = \frac{20}{5} = 4$$

Step 3: Q_C can be found by adding the value of Q_C in step 1 and step 2 = 1 + 1 = 2.

So, $Q_A : Q_B : Q_C = 1 : 4 : 2$

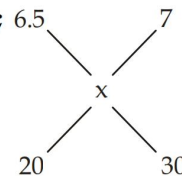
16. (a); Let average price = x per kg.



$$\therefore \frac{x-5}{6-x} = \frac{8}{4}$$

$$4x - 20 = 48 - 8x \Rightarrow 12x = 68 \Rightarrow x = 5.66$$

17. (b);



$$\frac{x-6.5}{7-x} = \frac{30}{20} \Rightarrow 2x - 13 = 21 - 3x$$

$$5x = 34 \Rightarrow x = 6.8 \text{ per kg.}$$

$\therefore$ Total value after mixture = $6.8 \times 50 = 340$ Rs.

Sold the mixture in $(340 + 60) = 400$ rs.

$$\therefore \text{required value per kg} = \frac{400}{50} = 8 \text{ per kg.}$$

18. (b); Water quantity in first case = $\frac{7}{20} \times 48$ litre

water quantity in second case = $\frac{17}{35} \times 42$ litre

Total water quantity in 90 litre of mixture

$$= \frac{7 \times 48}{20} + \frac{17 \times 42}{35} = \frac{186}{5} \text{ litre}$$

After adding of 20 litre of water the quantity will

$$\text{be} = \frac{186}{5} + 20 = \frac{286}{5}$$

quantity of wine in 110 litre of mixture

$$= 110 - \frac{286}{5} = \frac{264}{5}$$

$$\therefore \text{required ratio} = \frac{264}{5} : \frac{286}{5} = 12 : 13$$

19. (a); The quantity of milk in the $(2 + 5 + 9 = 16)$ ltr. of

$$\text{mixture} = \frac{90}{100} \times 2 + \frac{80}{100} \times 5 + \frac{70}{100} \times 9$$

$$= 1.8 + 4 + 6.3 = 12.1$$

$$\text{Water quantity} = 16.0 - 12.1 = 3.9$$

$$\therefore \text{Required Ratio} = \frac{121}{39} = 121 : 39$$

20. (c); Let y litre of mixture is released every time

$$\therefore \frac{12 \left(1 - \frac{y}{12}\right)^2}{12} = \frac{0.10}{0.40} \Rightarrow \left(1 - \frac{y}{12}\right)^2 = \frac{1}{4}$$

$$\left(1 - \frac{y}{12}\right)^2 = \left(\frac{1}{2}\right)^2 \Rightarrow \frac{y}{12} = \frac{1}{2} \Rightarrow y = 6 \text{ litres}$$

Previous Year (Memory Based)

1. (b); Let the weight of alloy is x kg.

$$x \times \frac{12}{100} = 69$$

$$x = 69 \times \frac{100}{12} = 575 \text{ kg}$$

2. (d);

	milk	water
Let we add x litre of	36	4
water	36	4 + x

$$\frac{4+x}{40+x} \times 100 = 20, \quad 20 + 5x = 40 + x$$

$$4x = 20, \quad x = 5 \text{ litres}$$

3. (b); Amount of alcohol in original solution

$$= \frac{15}{100} \times 400 = 60 \text{ ml}$$

Let we added x ml of pure alcohol to solution.

$$\frac{60+x}{400+x} \times 100 = 32$$

$$60 \times 100 + 100x = 32 \times 400 + 32x$$

$$100x - 32x = 32 \times 400 - 60 \times 100$$

$$68x = 400[32 - 15]$$

$$x = \frac{400 \times 17}{68} = 100 \text{ ml}$$

4. (a); Cost price of Tea after mixing = $\frac{100}{110} \times 68.2$
= 62 Rs./kg

60		65
	62	
65 - 62	:	62 - 60
= 3	:	= 2

required ratio = 3 : 2

5. (b); amount of gold in alloy = $\frac{80}{100} \times 50 = 40 \text{ gm}$

amount of Silver = 50 - 40 = 10 gm.

Let we added xgm of gold.

$$\frac{40+x}{50+x} \times 100 = 95$$

$$40 \times 100 + 100x = 95 \times 50 + 95x$$

$$100x - 95x = 4750 - 4000$$

$$5x = 750 \Rightarrow x = 150 \text{ gm.}$$

6. (b); While selling at Rs. 20/litres he got 25% profit

$$\text{Cost Price} = \frac{100}{125} \times 40 = 32 \text{ Rs. / kg}$$

Milk		Water
50		0
	32	
32 - 0	:	50 - 32
= 32	:	= 18

required ratio = 16 : 9

7. (b);
- | | |
|--------|---------|
| Type I | Type II |
| 280 | 240 |
| Av | |

$$\begin{array}{cc} \text{Av} - 240 & : & 280 - \text{Av} \\ 7\text{kg} & : & 9\text{kg} \end{array}$$

$$\frac{\text{Av} - 240}{280 - \text{Av}} = \frac{7}{9}$$

$$9\text{Av} - 2160 = 1960 - 7\text{Av}$$

$$16\text{Av} = 4120$$

$$\text{Av} = \frac{4120}{16} = 257.5 \text{ Rs./kg}$$

8. (b);
- | | |
|--------|---------|
| Type I | Type II |
| 30% | 50% |
| 45% | |

$$\begin{array}{cc} 50 - 45 & : & 45 - 30 \\ = 5 & : & = 15 \end{array}$$

$$\text{required ratio} = \frac{5}{15} = \frac{1}{3}$$

9. (a); Milk + Water = 40 litres

$$\begin{array}{cc} \frac{7}{8} \times 40 & \frac{1}{8} \times 40 \\ = 35 & = 5 \end{array}$$

$$\text{Let we added } x \text{ litres of water } \frac{35}{5+x} = \frac{3}{1}$$

$$35 = 3 \times 5 + 3 \times x, \quad 35 = 15 + 3x$$

$$3x = 20$$

$$x = \frac{20}{3} = 6\frac{2}{3}$$

10. (c); Let the volume of Jar is $9x$

	A	B
	$7x$	$2x$
Amount left after taking		
10 litre mixture out	$7x - 14$	$2x - 4$
After adding 10 litre of B	$7x - 14$	$2x - 4 + 18$

$$\frac{7x - 14}{2x + 14} = \frac{2}{3}$$

$$21x - 42 = 4x + 28$$

$$17x = 70, \quad x = \frac{70}{17}$$

$$\text{Initial amount of A} = 7x = 7 \times \frac{70}{17} = \frac{490}{17} \text{ litre}$$

11. (c);
- | | |
|------------------------|-------------------|
| milk | water |
| 20 | 10 |
| Let we added x litre | |
| of water | $20 \quad 10 + x$ |

$$\frac{20}{10+x} = \frac{1}{2}, \quad 40 = 10 + x, \quad x = 30 \text{ litre}$$

12. (c); amount of gold in A = $\frac{5}{8}$

$$\text{amount of gold in B} = \frac{5}{16}$$

$$\text{amount of gold in C} = \text{Av.}$$

$\frac{5}{8}$	$\frac{5}{16}$
Av	
$\text{Av} - \frac{5}{16}$	$\frac{5}{8} - \text{Av}$

$$\frac{\text{Av} - \frac{5}{16}}{\frac{5}{8} - \text{Av}} = \frac{1}{1}, \quad \text{Av} - \frac{5}{16} = \frac{5}{8} - \text{Av}$$

$$2\text{Av} = \frac{5}{8} + \frac{5}{16} = \frac{10+5}{16} = \frac{15}{16}, \quad \text{Av} = \frac{15}{32}$$

$$\text{amount of gold} = 1 - \frac{15}{32} = \frac{17}{32}$$

$$\text{The ratio of gold and copper} = \frac{15}{32} : \frac{17}{32} = 15 : 17$$

13. (b); Let 12 kg of both alloys are melted together

	Zinc	Copper	Tin	Lead
1 st Alloy	$12 \times \frac{2}{6} = 4 \text{ kg}$	6 kg	2 kg	—
2 nd Alloy	—	$\frac{5}{12} \times 12 = 5 \text{ kg}$	4 kg	3 kg
After melting	4 kg	11 kg	6 kg	3 kg

$$\text{The weight of lead per kg} = \frac{3}{4+11+6+3}$$

$$= \frac{3}{24} = \frac{1}{8} \text{ kg}$$

14. (a); The cost price of tea sold at 194.4 kg

$$= \frac{100}{120} \times 194.4 = 162 \text{ Rs/kg}$$

$$\begin{array}{ccc} 192 & & 150 \\ & \searrow & \nearrow \\ & 162 & \\ & \nearrow & \searrow \\ 162 - 150 & : & 192 - 162 \\ = 12 & : & = 30 \end{array}$$

$$\text{required ratio} = 12 : 30 = 2 : 5$$

15. (a); In first glass amount of milk = $\frac{3}{8}$

$$\text{In second glass amount of milk} = \frac{6}{7}$$

$$\text{The amount of milk in resultant mixture} = \frac{1}{2}$$

$$\begin{array}{ccc} \frac{3}{8} & & \frac{6}{7} \\ & \searrow & \nearrow \\ & \frac{1}{2} & \\ & \nearrow & \searrow \\ \frac{6}{7} - \frac{1}{2} & & \frac{1}{2} - \frac{3}{8} \\ = \frac{12-7}{14} & & = \frac{4-3}{8} \\ = \frac{5}{14} & & = \frac{1}{8} \end{array}$$

$$\text{required ratio} = \frac{\frac{5}{14}}{\frac{1}{8}} = \frac{5}{14} \times \frac{8}{1} = \frac{20}{7} = 20 : 7$$

16. (d); Cost price of mixture = $\frac{100}{120} \times 324 = 270 \text{ Rs / kg}$

$$\begin{array}{ccc} \text{Darjeeling Tea} & & \text{Assam Tea} \\ 320 & & 250 \\ & \searrow & \nearrow \\ & 270 & \\ & \nearrow & \searrow \\ 270 - 250 & : & 320 - 270 \\ = 20 & : & = 50 \\ \frac{\text{Darjeeling Tea}}{\text{Assam Tea}} = \frac{20}{50} = \frac{2}{5} \end{array}$$

17. (b); water(1) + glycerine(3) = 240cc
60cc 180cc

Let we add x cc water in the mixture

$$\frac{60+x}{180} = \frac{2}{3}, \quad 180 + 3x = 360$$

$$3x = 360 - 180 = 180, \quad x = 60 \text{ cc.}$$

18. (b); amount of acid in A = $\frac{4}{7}$

$$\text{amount of acid in B} = \frac{5}{8}$$

$$\text{amount of acid in C} = \frac{3}{5}$$

$$\begin{array}{ccc} A & & B \\ \frac{4}{7} & & \frac{5}{8} \\ & \searrow & \nearrow \\ & C & \\ & \frac{3}{5} & \\ & \nearrow & \searrow \\ \frac{5}{8} - \frac{3}{5} & : & \frac{3}{5} - \frac{4}{7} \\ = \frac{25-24}{40} & : & = \frac{21-20}{35} \\ = \frac{1}{40} & : & = \frac{1}{35} \\ \frac{A}{B} = \frac{\frac{1}{40}}{\frac{1}{35}} = \frac{35}{40} = \frac{7}{8} \end{array}$$

19. (a); Amount of acid in A = $\frac{5}{7}$

$$\text{Amount of acid in B} = \frac{8}{13}$$

$$\text{Amount of acid in resultant mixture} = \frac{9}{13}$$

$$\begin{array}{ccc} A & & B \\ \frac{5}{7} & & \frac{8}{13} \\ & \searrow & \nearrow \\ & \frac{9}{13} & \\ & \nearrow & \searrow \\ \frac{9}{13} - \frac{8}{13} & : & \frac{5}{7} - \frac{9}{13} \\ = \frac{1}{13} & : & = \frac{2}{91} \\ \frac{A}{B} = \frac{\frac{1}{13}}{\frac{2}{91}} = \frac{91}{2 \times 13} = \frac{7}{2} \end{array}$$

20. (c);	sprit	water
mixture of 20 litres	6	14
mixture of 36 litres	21	15
	27	29

required ratio = 27 : 29

21. (c); Amount of steel in Ist type = $\frac{11}{13}$

Amount of steel in IInd type = $\frac{21}{26}$

Amount of steel in resultant mixture = $\frac{32}{39}$

$$\begin{array}{ccc}
 \text{I}^{\text{st}} \text{ type} & & \text{II}^{\text{nd}} \text{ type} \\
 \frac{11}{13} & & \frac{21}{26} \\
 & \swarrow \quad \searrow & \\
 & \frac{32}{39} & \\
 & \swarrow \quad \searrow & \\
 \frac{32}{39} - \frac{21}{26} & : & \frac{11}{13} - \frac{32}{39} \\
 = \frac{64 - 63}{78} & : & = \frac{33 - 32}{39} \\
 = \frac{1}{78} & : & = \frac{1}{39} \\
 \frac{\text{I}^{\text{st}} \text{ type}}{\text{II}^{\text{nd}} \text{ type}} = \frac{\frac{1}{78}}{\frac{1}{39}} = \frac{39}{78} = \frac{1}{2}
 \end{array}$$

22. (d); Amount of gold in A = $\frac{7}{9}$

Amount of gold in B = $\frac{7}{18}$

Let the amount of gold in C is Avg.

$$\begin{array}{ccc}
 \frac{7}{9} & & \frac{7}{18} \\
 & \swarrow \quad \searrow & \\
 & \text{Avg} & \\
 & \swarrow \quad \searrow & \\
 \text{Avg} - \frac{7}{18} & & \frac{7}{9} - \text{Avg} \\
 \frac{\text{Avg} - \frac{7}{18}}{\frac{7}{9} - \text{Avg}} = \frac{1}{1}
 \end{array}$$

$$\text{Avg} - \frac{7}{18} = \frac{7}{9} - \text{Avg}$$

$$2\text{Avg} = \frac{7}{9} + \frac{7}{18} = \frac{14+7}{18} = \frac{21}{18} = \frac{7}{6}$$

$$\text{Avg} = \frac{7}{12}$$

$$\text{Amount of copper} = 1 - \frac{7}{12} = \frac{5}{12}$$

$$\text{ratio of gold to copper} = \frac{7}{12} : \frac{5}{12} = 7 : 5$$

23. (c); Let be volume of can is

$$\begin{array}{ccc}
 & 48v & \\
 & \swarrow \quad \searrow & \\
 28v & & 20v \\
 \text{A} & & \text{B}
 \end{array}$$

After operation ratio of A and B is 7 : 9

	A	B
	28v	20v
Amount left After taking out 9L	$28v - \frac{21}{4}$	$20v - \frac{15}{4}$
After adding 9L of B	$28v - \frac{21}{4}$	$20v - \frac{15}{4} + 9$

$$\frac{28v - \frac{21}{4}}{20v - \frac{15}{4} + 9} = \frac{7}{9}, \quad v = \frac{3}{4}$$

$$\text{Initial Amount of A} = \frac{3}{4} \times 48 \times \frac{7}{12} = 21\text{L}$$

24. (d); 29 Rs./kg. 20 Rs./kg.

$$\begin{array}{ccc}
 & & \\
 & \swarrow \quad \searrow & \\
 & \text{Avg} & \\
 & \swarrow \quad \searrow & \\
 \text{Avg} - 20 & & 29 - \text{Avg} \\
 15 \text{ kg} & & 25 \text{ kg}
 \end{array}$$

$$\frac{\text{Avg} - 20}{29 - \text{Avg}} = \frac{15}{25}, \quad \frac{\text{Avg} - 20}{29 - \text{Avg}} = \frac{3}{5}$$

$$5 \text{ Avg} - 100 = 87 - 3 \text{ Avg}$$

$$8 \text{ Avg} = 187$$

$$\text{Avg} = \frac{187}{8} = 23.375$$

$$\text{profit} = 27 - 23.375 = 3.625$$

$$\text{profit for whole transaction} = 3.625 \times (15 + 25)$$

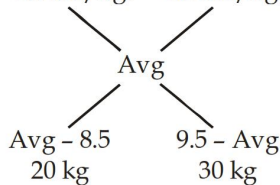
$$= 3.625 \times 40 = \text{Rs. } 145$$

25. (d); Let the volume of each mixture is $120x$

	acid	water
1 st mixture	$80x$	$40x$
2 nd mixture	$72x$	$48x$
3 rd mixture	$75x$	$45x$
	$227x$	$133x$

The ratio of water and acid in the resultant mixture = $133 : 227$

26. (d); 9.5 Rs./kg. 8.5 Rs./kg.



$$\frac{\text{Avg} - 8.5}{9.5 - \text{Avg}} = \frac{20}{30}$$

$3 \text{ Avg} - 25.5 = 19 - 2 \text{ Avg}$
 $5 \text{ Avg} = 44.5, \text{ Avg} = 8.9 \text{ Rs./kg}$
 So, no profit no loss

27. (b); Price of pure milk = 3 per litre
 Price of mixture = 1.5 per litre
 So in mixture of 1 litre 500 ml will be pure milk and 500 ml will be water
 milk : water = 1 : 1
 amount of milk in mixture = amount of water = 16 litres.

28. (d); Let the capacity of vessels are $21v, 14v, 7v$

we take $\frac{1}{3}$ of first, $\frac{1}{2}$ of second and $\frac{1}{7}$ of third and mix it amount taken from first mixture

$$= \frac{1}{3} \times 21v = 7v$$

$$\text{second mixture} = \frac{1}{2} \times 14v = 7v$$

$$\text{third mixture} = \frac{1}{7} \times 7v = v$$

	milk	water
first ($7v$) (milk : water = 5 : 2)	$5v$	$2v$
second ($7v$) (milk : water = 4 : 1)	$\frac{4}{5} \times 7v$	$\frac{1}{5} \times 7v$
third (v) (milk : water = 4 : 1)	$\frac{4}{5} \times v$	$\frac{1}{5} \times v$
mixture ($15v$)	$11.4v$	$3.6v$

$$\text{percentage of water} = \frac{3.6v}{15v} \times 100 = 24\%$$

29. (c); mixture (2kg) copper aluminium
 0.6kg 1.4kg

Let we added x kg aluminium powder

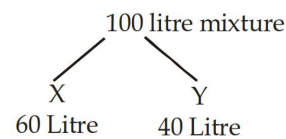
$$\frac{0.6}{2+x} \times 100 = 20$$

$$60 = 40 + 20x, \quad 20 = 20x$$

$$x = 1 \text{ kg} = 1000 \text{ gm}$$

30. (b); cost price of liquid = $\frac{100}{110} \times 11 = 10$ Rs per litre

Let we take 100 litre of mixture



Price liquid Y is a Rs per litre so price of X will be $a + 2$ Rs per litre

$$100 \times 10 = 60 \times (a + 2) + 40 \times (a)$$

$$1000 = 60a + 120 + 40a$$

$$880 = 100a$$

$$a = 8.8$$

$$\text{cost price of X} = 8.8 + 2 = 10.8 \text{ Rs./litre}$$